

Gyu Seon Kim

Ph.D., Aerospace AI Research Engineer

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Profile

- Ph.D., Aerospace AI Research Engineer at the R&D Center, RealtimeVisual Co., Ltd., Seoul, Republic of Korea. Member of the IEEE.

Research Interests

- **Autonomous Aerospace Systems:** Control and Communication for Autonomous Aerospace Systems (e.g., Aircraft, Drone, UAV, UAM, Fixed Wing, Rotary Wing, Rotorcraft, Launch Vehicles, and Spacecraft)
- **AI for Defense and Military Applications:** Intelligent Control and Autonomous Decision-Making for Defense and Aerospace Systems (e.g., Fighter Aircraft, Military Drone, Anti Drone System, Missile, Decoy, and Torpedo)
- **MARL-based Autonomous Control:** Multi-Agent Reinforcement Learning for Cooperative and Networked Aerial Vehicle Systems
- **RL-based Satellite Communication Systems:** Reinforcement Learning-based Communication and Networking for Satellite Systems (e.g., LEO Satellite Routing, Cube Satellite Design, Satellite Constellations, and Space-Air-Ground Integrated Networks)
- **Quantum AI:** Quantum Reinforcement Learning for Aerospace Control and Communication Systems

Education

Ph.D. Korea University - College of Engineering, Electrical and Computer Engineering Seoul, Republic of Korea (ECE)

B.S. Inha University - College of Engineering, Aerospace Engineering Incheon, Republic of Korea

Academic Background

- **Korea University - Research Institute of Information and Communication Technology**, Seoul, Republic of Korea - *Postdoctoral Researcher*
- **University of Houston - College of Engineering (ECE)**, Houston, TX, USA - *Visiting Scholar, Wireless Networking, Signal Processing and Security Laboratory (Prof. Zhu Han)*
- **Korea University - College of Engineering (ECE)**, Seoul, Republic of Korea - *Research Assistant, Artificial Intelligence and Mobility Laboratory (Advisor: Prof. Joongheon Kim)*

Industry Background

- **RealtimeVisual - R&D Center**, Seoul, Republic of Korea - *Aerospace AI Research Engineer (Military Service Exception)*

Awards and Honors

- **IEIE Best Paper Award** (November 2025) - *Lightweight Control Modeling via Quantum Reinforcement Learning*
- **IPESK Next Generation Engineering Researcher Award** (October 2025) - Funded by an Institute for Promotion of Engineering and Science of Korea
- **IEEE Vehicular Technology Society Student Scholarship Award** (October 2025) - Funded by an IEEE VTS Endowment Fund and partially funded by a VTS Award Fund, and administered by the IEEE Foundation
- **KICS Best Paper Award** (November 2024) - *Reinforcement Learning-Based Counter Measure Tactics to Avoid Torpedo Threats*
- **IEEE Seoul Section Student Paper Contest, Best Paper Award** (December 2023) - *Aircraft Taxi Routing using Reinforcement Learning at Hartsfield Jackson Atlanta International Airport*
- **KICS Best Paper Award** (February 2023) - *Self-Learning-Based Hybrid MAC for Military UAV Networks*

Review Experience

- IEEE Transactions on Aerospace and Electronic Systems
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Mobile Computing
- IEEE Transactions on Communications
- IEEE Transactions on Industrial Informatics
- IEEE Transactions on Network and Service Management
- IEEE Transactions on Cognitive Communications and Networking
- IEEE Internet of Things Journal
- IEEE Communications Magazine
- IEEE Access
- IEEE/KICS Journal of Communications and Networks
- IEEE/KICS ICT Express
- Springer Nature Aerospace Systems
- ACM Computing Surveys
- KSII Transactions on Internet and Information Systems
- ETRI Journal

Teaching Assistants

- **SW Programming Basics (GECT002), Dept. of ECE, Korea University** - Spring 2025

Professional Profiles

- Academic email: kingdom0545@korea.ac.kr
- Laboratory: [Artificial Intelligence and Mobility Laboratory](#)
- [Google Scholar](#)
- [LinkedIn](#)
- [DBLP](#)
- [DBpia](#)
- [GitHub](#)
- [IEEE Xplore author profile](#)

Professional References

- **Prof. Joongheon Kim**, Ph.D. Research and Dissertation Advisor - Professor, Korea University, Seoul, Republic of Korea; joongheon@korea.ac.kr [Website](#)
- **Prof. Zhu Han**, Ph.D. Visiting Research Supervisor - Professor, University of Houston, Houston, TX, USA; zhan2@uh.edu [Laboratory website](#)